Taeyoung Wang

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RESEARCH INTERESTS

Electrolyte chemistry and materials design for electrochemical energy storage, particularly how solvation, phase behavior, and ion transport govern battery performance.

EDUCATION

Columbia University Graduation Date: May 2027
- B.S. in Materials Science & Engineering Cumulative GPA: 3.94/4.00
- **Honors:** Dean's List (all semesters); Summer@SEAS Undergraduate Research Fellowship, 2026 ($6,000 stipend)

University of Illinois at Urbana-Champaign 2023 — 2024
- Undergraduate studies in Chemistry Cumulative GPA: 3.92/4.00
- Liberal Art and Science College (LAS) James Scholar Honor Program

PUBLICATIONS & MANUSCRIPTS

1. **Wang, T.**, et al. "Probing Salt Partitioning and Low-Temperature Transport in Gel Polymer Electrolytes" (Manuscript in preparation)

2. Yang, Z.*; Milazzo, T.*; Bao, W.; **Wang, T.**, et al. "Room-temperature K-S Flow Battery Enabled by K_2S_2/K_2S-soluble Formamide Electrolyte for Long-duration Energy Storage." (Submitted, 2026)

   *These authors contributed equally to this work.

3. Bao, W.; Yang, Z.; **Wang, T.**, et al. "Economic Analysis of Light Isotope-enriched Elements for Aerospace Applications." (Submitted, 2026)

RESEARCH EXPERIENCE

Quantifying Self-Concentration Polarization in Heterogeneous Li-Ion Battery Electrolytes 2026 — Present
Research Assistant
- Investigated equilibrium salt and solvent partitioning across battery components and gel–liquid phases using quantitative NMR and Raman spectroscopy
- Developed a quantitative $^{1}H/^{19}F$ NMR workflow with sample-specific relaxation measurements to assess electrolyte composition while addressing relaxation-induced quantification bias
- Identified changes in solvent composition and salt-anion ratios during gel curing; refined the curing protocol to reduce compositional changes that confound partitioning measurements
- Evaluated liquid and gel electrolyte transport using low-temperature impedance measurements and investigated contact-resistance and cell-fabrication artifacts affecting conductivity comparisons

K-S Batteries with K-β"-Alumina Solid Electrolytes 2025 — Present
Research Assistant
- Assembled and tested symmetric and full CR2032 cells for room-temperature K–S batteries with β''-alumina solid electrolytes; prepared K–Na alloy anodes and polysulfide catholytes in formamide and TEGDME
- Quantified potassium-polysulfide solubility across candidate solvents by stepwise titration, identifying formamide as a high-solubility catholyte solvent with a K_2S solubility of 2.75 $\mathrm{mol\,kg^{-1}}$
- Assembled and operated a semi-flow prototype with circulating formamide catholyte and a K–Na alloy anode, demonstrating cycling over 70 cycles and more than 400 hours

– Calculated theoretical energy density and evaluated levelized cost of storage and installation costs to assess the system's potential for long-duration energy storage

Solubility Measurements for Machine-Learning-Assisted Electrolyte Screening 2025 — Present
Research Assistant

– Measured K_2S solubility across solvents and solvent mixtures, providing experimental data for machine-learning screening to identify candidates with high K_2S solubility.

Techno-Economic Modeling of Light Isotope Enriched Elements for Aerospace Lightweighing 2025
Research Assistant

– Implemented the techno-economic model quantifying the benefit of replacing structural aerospace materials with versions enriched in their lightest stable isotope, computing per-element economic benefit across elements
– Calculated enrichment cost from separative work unit (SWU) analysis of multi-stage separation cascades, deriving cost-per-SWU scaling factors for gas centrifugation and chemical exchange benchmarked against ^{235}U and deuterium
– Optimized the light-isotope enrichment level and stripping stage ratio for each element to maximize economic benefit while constraining feed-material consumption

Research on Redox Flow Batteries & Automation and High Throughput Electrochemistry 2024
Research Assistant

– Worked at Joaquín Rodríguez-López (JRL) Lab on solid state redox flow batteries and the automation of electrochemical experiments

POSTERS

Wang, T., et al. "Probing Salt Partitioning and Low-Temperature Transport in Gel Polymer Electrolytes." *Columbia Undergraduate Research Symposium*, New York, USA (Accepted poster; presentation scheduled October 2026)

ADDITIONAL EXPERIENCE

Samsung Electronics DS Shining Star Program 2024
Selected Participant

– Studied core semiconductor manufacturing processes and device physics and gained
– Gained first-hand exposure to semiconductor mass production through fab tours and technical mentoring with practicing engineers

Beckman Open House 2023 — 2024
Presentor

– Planned and performed exhibition for electrochemistry and batteries, collaborating with the JRL Lab's electrochemistry robot

TECHNICAL SKILLS

Electrochemistry: Coin & pouch cell assembly and disassembly; symmetric and half-cell testing; catholyte preparation; LFP electrode fabrication; electrode calendering
Materials Characterization: XRD; SEM; Quantitative NMR; Raman Spectroscopy; Optical Microscopy
Data Analysis: Origin; Cyclic Voltammetry diagram analysis; Electrochemical Impedance Spectrscopy analysis
Laboratory Techniques: Glovebox handling; Preparation for XANES and ICP